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Q4 & Q5 Assignment 1: is the grader correct? @STAFF please check

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32 minutes ago

Both Q4 and Q5 seem to be pretty straight-forward but grader does not accept my reasoning:

Q4:

p := the specific person in population with 100 people

E_1 := event for selecting p from population in the **first** SRS **S_1** of size 20

E_2 := event for selecting p from population in the **second** SRS **S_2** of size 20

$$P(E_1) = P(p \in S_1) = 1 - P(p \notin S_1) = 1 - \frac{\binom{99}{20}}{\binom{100}{20}}$$

$$= 1 - \frac{99!20!80!}{100!20!79!} = 1 - \frac{80}{100} = 1 - \frac{4}{5} = \frac{1}{5}$$

Similarly,

$$P(E_2) = P(p \in S_2) = \frac{1}{5}$$

Now, E_1 and E_2 independent, hence

$$P(E_1 \cap E_2) = P(E_1)P(E_2) = \frac{1}{5} \cdot \frac{1}{5} = \frac{1}{25}$$

But grader does not accept.

Q5:

Population (**P**) size: 6

Sample (**S**) size: 2

E_1 := event that individual **a** is selected from population in first extraction for the sample **S**

E_2 := event that individual **a** is not selected from population in first extraction but selected in the second extraction for the sample **S**

Note that E_1 and E_2 are mutually exclusive here.

Hence,

$$P(E_1) = \frac{1}{6}$$

$$P(E_2) = P(a \notin \text{first draw}) \cdot P(a \in \text{second draw}) = \left(1 - \frac{1}{6}\right) \cdot \frac{1}{5} = \frac{5}{6} \cdot \frac{1}{5} = \frac{1}{6}, \text{ assuming SRSWOR (without replacement)}$$

Hence, probability that **a** is selected throughout the sampling process is:

$$P(E_1 \cup E_2) = P(E_1) + P(E_2) = \frac{1}{6} + \frac{1}{6} = \frac{1}{3} \text{ (by countable additivity and mutual exclusion)}$$

But again the grader does not accept.

Not sure where I am wrong for Q4 and Q5.

@STAFF, can you please help and double check if the grader is correct? Thanks in advance.